Rachel Thomas

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Technical Skills

Languages: Rust, C, C++, Python, Java, TypeScript/JavaScript, Assembly (x86/ARM), Swift, SQL

Education

University of Maryland at College Park Aug 2024 – Present
Doctor of Philosophy in Computer Science | Dean's Fellowship *College Park, MD*

University of North Carolina at Chapel Hill Jan 2023 – May 2024
Master of Science in Computer Science *Chapel Hill, NC*

University of North Carolina at Chapel Hill Aug 2018 – Dec 2022
Bachelor of Science in Computer Science | Magna Cum Laude *Chapel Hill, NC*

- Minor in Mathematics

Experience

Graduate Assistant Aug 2024 – Present
Department of Computer Science, University of Maryland *College Park, MD*

- Supported instruction for CMSC 414: *Computer and Network Security* and CMSC 839C: *Governing Algorithms & Algorithmic Governance*, providing technical guidance, mentorship, grading, and course support.
- Managed course logistics, grading, and resources while holding office hours and providing technical guidance and mentorship to students on topics including cryptography, network security, and algorithmic governance.
- Conducted research in applied cryptography and human-centered security, collaborating with interdisciplinary teams to design and implement privacy-preserving cryptographic protocols.
- Developed cryptographic constructions and software prototypes through independent and collaborative research, contributing to ongoing research projects and publications.

Research Assistant Aug 2022 – May 2024
Department of Computer Science, UNC Chapel Hill *Chapel Hill, NC*

- Conducted research in privacy-enhancing technologies, cryptography, and side-channel attacks, collaborating with researchers to develop novel cryptographic techniques. Co-authored peer-reviewed publications and presented findings at academic conferences.
- Assisted in mentoring undergraduate students and junior researchers, guiding them through complex research methodologies and experimental designs.

Learning Assistant Aug 2021 – Dec 2021
Department of Mathematics, UNC Chapel Hill *Chapel Hill, NC*

- Supported instruction for MATH 233: *Calculus of Functions of Several Variables*, assisting with course delivery, student engagement, and classroom activities.
- Led individual and group tutoring sessions during office hours, helping students develop proficiency in multivariable calculus concepts and problem-solving techniques.

Publications & Preprints

CheckOut: User-Controlled Anonymization for Customer Loyalty Programs July 2024
Matthew Gregoire, ***Rachel Thomas****, Saba Eskandarian* *PoPETS 2024*

- Published in the *24th Privacy Enhancing Technologies Symposium (PoPETS)*

Cryptographic Personas: Responsible Pseudonyms Without De-Anonymization October 2025
Rachel Thomas*, Oliwia Kempinski, Hari Kailad, Emma Margaret Shroyer, Ian Miers, Gabriel Kaptchuk* *Preprint*

- Preprint available through the *IACR Cryptology ePrint Archive*, Paper 2025/1969

A Cryptographic Perspective on California's Delete Request and Opt-out Platform July 2026
Aarushi Goel, Gabriel Kaptchuk, Peihan Miao, Phuoc Van Long Pham, Satvinder Singh, ***Rachel E. Thomas*** *Preprint*

- Preprint available through the *IACR Cryptology ePrint Archive*, Paper 2026/1421

Projects

Zero-Knowledge Social Proofs June 2025 – Present

Department of Computer Science at University of Maryland *College Park, MD*

- Designed a privacy-preserving proof-of-personhood framework using zero-knowledge proofs to achieve Sybil resistance and unlinkable identity verification without revealing user attributes.
- Leveraged zkTLS-based attestations and exploring OPRF- and VDF-based constructions to enable secure account ownership proofs, deduplication, and abuse resistance under adversarial conditions.

A Cryptographic Perspective on California's DROP June 2025 – Present

Department of Computer Science at University of Maryland *College Park, MD*

- Conducted a cryptographic analysis of California's Delete Request and Opt-out Platform (DROP), identifying opportunities to strengthen privacy, accountability, and verifiable processing guarantees for statewide data deletion requests.
- Designed and evaluated privacy-preserving protocol enhancements using Private Set Intersection (PSI), zero-knowledge proofs, and authenticated commitments to improve the security and auditability of large-scale data deletion workflows.

Cryptographic Personas: Responsible Pseudonyms without De-Anonymization Aug 2024 – Present

Department of Computer Science at University of Maryland *College Park, MD*

- Developed *cryptographic personas*, a privacy-preserving messaging framework enabling unlinkable pseudonymous identities with policy-based, revocable anonymity for moderation without requiring de-anonymization.
- Optimized efficiency via offline zero-knowledge proof generation and folding techniques, and integrated prototype deployments with messaging platforms (Signal, Slack) for end-to-end evaluation in real-time settings.

Side-Channel Analysis of Microsoft Edge Password Leak Detection Protocol Aug 2023 – May 2024

Department of Computer Science at UNC Chapel Hill *Chapel Hill, NC*

- Analyzed side-channel leakage in Microsoft Edge's Password Leak Detection pipeline and Microsoft SEAL homomorphic encryption library, identifying information leakage vectors under realistic adversarial assumptions.
- Developed and evaluated practical attack methodologies across native and browser-based environments, constructing threat models to assess protocol robustness against timing and microarchitectural leakage.

CheckOut: User-Controlled Anonymization for Customer Loyalty Programs May 2022 – May 2024

Department of Computer Science at UNC Chapel Hill *Chapel Hill, NC*

- Designed and implemented *CheckOut*, a privacy-preserving loyalty system enabling unlinkable transaction authorization via user-controlled pseudonymization of loyalty identifiers, preventing cross-merchant tracking.
- Developed a cryptographic protocol and mobile prototype leveraging randomized barcode representations and lightweight client-side transformations to enforce selective disclosure of loyalty credentials while preserving reward correctness and auditability.